

Beyond Abundance: Pollinator Functional Trait Composition and Temporal Visitation Patterns as Predictors of Pollination Success in Urban Home Gardens

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Abstract

Total pollinator abundance is a critical predictor of plant reproductive success, but less empirical analysis has been conducted on how pollinator functional traits and visitation patterns impact pollination outcomes. This paper examines how community-weighted mean (CWM) bee tongue length and body size, as well as the temporal distribution of pollinator visits across the day, predict garden-level pollination success independently of abundance, and whether these associations vary with flower morphology. Using an open dataset of 5,794 individually measured pollinators from a standardised field experiment across 24 home gardens in Zurich, Switzerland (Reji Chacko et al., 2025), I constructed a garden-level master dataset linking CWM bee traits and hourly temporal visitation patterns to seed and fruit set outcomes for four phytometer plant species. CWM bee tongue length significantly predicted comfrey (*Symphytum officinale*) fruit set ($r = +0.45$, $p = 0.029$), consistent with trait-matching theory for this deeply euphilous flower. Bee CWM body size showed an exploratory negative association with radish (*Raphanus sativus*) fruit set ($r = -0.47$, $p = 0.020$). Morning visit fraction was the strongest single predictor found, explaining 31% of variance in radish fruit set ($r = +0.56$, $p = 0.004$). No trait or temporal predictor was significant for wild carrot (*Daucus carota*), consistent with its open generalist flower. These findings suggest that pollinator identity — who visits and when — matters beyond visit count abundance, and that trait-matching effects are plant-specific.

AI use: Claude Opus and ChatGPT 5.5 were used to assist with data analysis, including dataset cleaning and graph visualisation in Python.

Keywords: pollination ecology; trait-matching; community-weighted mean; urban gardens; temporal foraging; flower morphology

1. Introduction and Research Objectives

Urban densification, through the conversion of green space to impervious surface, reduces the availability of nesting substrates and floral resources, thereby restructuring pollinator communities (Liang et al., 2023; Wenzel et al., 2020). These environmental changes disproportionately disadvantage morphological specialists — species whose functional traits, such as tongue length and body size, are closely matched to the structural requirements of particular flower types — while favouring generalist species such as honeybees (*Apis mellifera*), which can exploit a broader range of floral resources across fragmented habitats. As specialist pollinators decline, plants dependent on trait-matched visitors experience reduced pollination quality, with cascading consequences for both wild plant reproductive success and agricultural pollination services (Harrison & Winfree, 2015).

This dynamic demands more quantitative research on trait-matching: the idea that effective pollination depends not only on the quantity of visitors, but on the morphological match between pollinator functional traits and flower structure (Armbruster, 2017; Rosas-Guerrero et al., 2014). Tongue length determines which flower morphologies a bee can access: long-tongued bees can reach deeply concealed nectar in euphilous flowers such as comfrey (*Symphytum officinale*), while short-tongued generalists cannot access the reward without bypassing the reproductive parts through nectar robbery. Body size, a correlated trait, scales with foraging range and influences which gardens pollinators can reach in fragmented urban landscapes (Greenleaf et al., 2007).

Reji Chacko et al. (2026) directly addressed this system, demonstrating across 24 home gardens in Zurich, Switzerland that urban densification negatively affects pollinator visitation frequency and richness, and that local floral richness can partially compensate for pollination success loss in specialist plants. Their study measured individual-level body size and tongue length, but analysed these traits as responses to environmental gradients (floral richness, impervious surface cover) rather than as predictors of plant reproductive success. Whether the functional composition of pollinator communities — specifically their community-weighted mean (CWM) body size and tongue length — explains variance in seed and fruit set beyond what pollinator group abundance already predicts was not addressed.

The same dataset additionally offers hourly visitation records across a nine-hour daily sampling window (09:00–18:00), yet the temporal distribution of visits has not been explored as a predictor of pollination outcomes. Floral receptivity varies across the day: stigma receptivity and pollen viability frequently peak in morning hours, particularly in many species with diurnal anthesis patterns (Primack, 1985). Gardens in which most pollinator activity is concentrated during periods of peak receptivity may therefore achieve higher fertilisation rates than those with equivalent total visit counts but a different activity distribution. I expand the analysis of Reji Chacko et al. (2025) in both pollinator trait composition and temporal resolution, addressing a complementary set of research questions:

RQ1: Does CWM bee tongue length predict garden-level pollination success, and is this effect stronger for plants with more specialised flower morphologies?

RQ2: Does CWM bee body size predict pollination success, and in which direction and for which plant species?

RQ3: Does the temporal distribution of pollinator visits — specifically the proportion occurring in morning hours — predict pollination success independently of total visit abundance?

Together, these questions test pollinator traits and temporal visit patterns as predictors of plant reproductive success, a complementary perspective to the environmental-response framing adopted by Reji Chacko et al. (2026), using the same openly available dataset.

2. Methods

2.1 Dataset and Study Design

I use the open dataset published by Reji Chacko, Moretti, and Frey (2025) in *Data in Brief* (Vol. 62, DOI: 10.1016/j.dib.2025.112013), archived on the EnviDat repository (DOI: 10.16904/envidat.676, CC-BY 4.0). The dataset originates from a standardised field experiment across 24 privately owned home gardens in Zurich, Switzerland during summer 2016. Gardens were selected to vary independently in local flowering plant species richness and landscape-scale urban density (proportion of impervious surface in a 500-m radius), following a quasi-orthogonal design.

In each garden, arrays of 19 experimental pots containing four insect-pollinated phytometer species were installed: wild carrot (*Daucus carota* L.), radish (*Raphanus sativus* L.), sainfoin (*Onobrychis viciifolia* Scop.), and comfrey (*Symphytum officinale* L.). These four species span a gradient of floral specialisation (Corbert, 2006): carrot has fully exposed nectar (halophilous); radish has partly concealed nectar (hemiphilous); sainfoin and comfrey have progressively more concealed nectar (euphilous). Pollinators were sampled by pre-trained volunteer scientists in 9-hour daily observation rounds (09:00–18:00). Each captured individual was identified to species level and measured for body size and tongue length.

2.2 Master Dataset Construction

I constructed a garden-level master table (96 rows: 24 gardens × 4 plants) from the downloaded source files. AI tools including Claude Opus and ChatGPT 5.5 were used to clean, merge, and cross-reference datasets in Python.

Pollination success outcomes. For carrot, mean seed counts per umbel were averaged to the garden level. For the other three species, fruit set proportion (flowers producing fruit / total flowers assessed) was aggregated from branch to plant to garden level.

Abundance variables. I summed the temporal visitation matrix to season totals per garden × plant, computing total visits, species richness, and group-level counts (bees, hoverflies, wasps, beetles). I further disaggregated bee visits by sociality using the `pollinator_group_sociality` field: honeybee visits (*Apis mellifera*), social bee visits (mainly *Bombus* spp.), and solitary bee visits. I constructed a `focal_group_visits_benchmark` variable per plant matching the JAE paper's Figure S6 predictors (hoverfly visits for carrot; social bee visits for radish and comfrey; total bee visits for sainfoin).

Community-weighted mean (CWM) traits. Using `individual_traits.csv`, I applied two cleaning steps: (1) values of 0 in trait columns were replaced with NaN (0 indicates an unmeasured trait); (2) nectar robbers (4.4% of captured individuals) were excluded from CWM calculations, as they did not provide legitimate pollination. I then computed arithmetic means of bee tongue length (proboscis length, mm) and bee body size (intertegular distance, mm) per garden × plant.

Temporal variables. I aggregated visits to total per hourly window per garden × plant, grouped into morning (09:00–12:00), midday (12:00–15:00), and afternoon (15:00–19:00) bins, computing the proportion of daily visits in each bin. Temporal evenness was computed

as Shannon entropy $H / \log(n \text{ windows observed})$.

2.3 Statistical Analysis

I used Pearson correlation to test pairwise associations between each predictor and pollination success, separately for each plant species ($n = 24$ gardens per plant). I report Pearson r , two-tailed p -value, and sample size for each association, using conventional thresholds: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), and $p < 0.10$ noted as marginal (.). All results are treated as exploratory given the small sample size. R^2 (Pearson r squared) is reported in Table 1 as a standardised measure of variance explained.

This analysis operates at garden level, sacrificing the repeated-measures structure exploited by the original paper's Bayesian GLMMs. I do not control for landscape-scale densification or local floral richness, both of which the original paper identified as significant drivers of abundance. Results should be treated as hypothesis-generating with known limitations.

3. Results

3.1 Pollination Success Varies Substantially Across Gardens

Pollination success showed substantial garden-to-garden variation for all four plant species (Figure 1). Comfrey fruit set ranged from 0.08 to 0.72 (mean = 0.50, SD = 0.16), reflecting high sensitivity to visitor identity. Sainfoin had the lowest mean fruit set (mean = 0.16, SD = 0.10) and highest proportional variance, partly due to early flowering synchrony issues. Radish showed low variance (mean = 0.28, SD = 0.05). Carrot seed set ranged from 81 to 278 seeds per umbel (mean = 174.5, SD = 39.9).

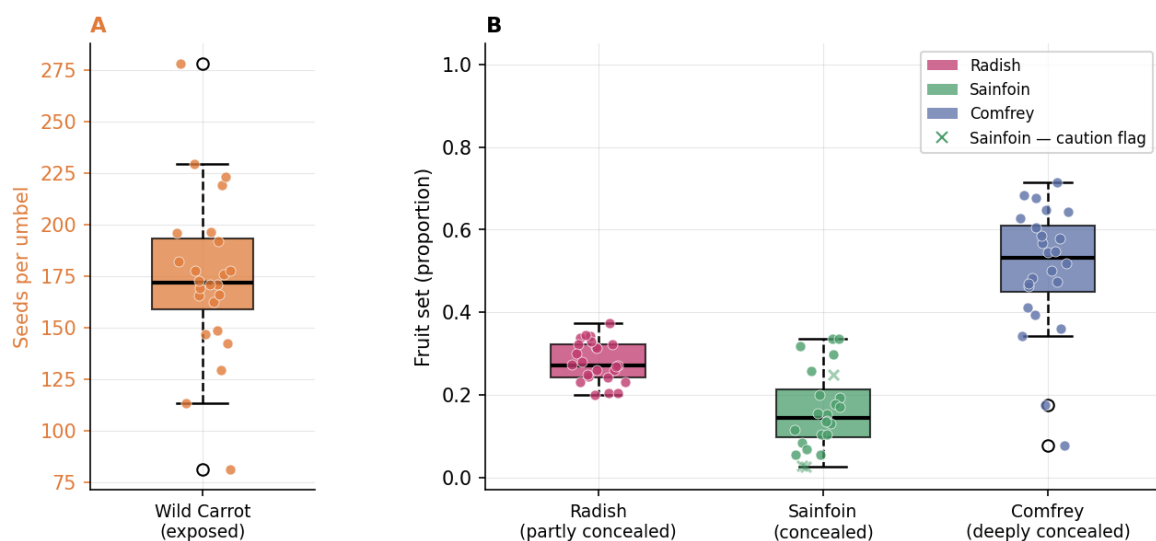


Figure 1. Pollination success by plant species. (A) Wild carrot: seeds per umbel (natural scale, separate axis). (B) Radish, sainfoin, and comfrey: fruit set proportion (0–1). Each point = one garden ($n = 24$). $\times$ = sainfoin caution-flagged gardens.

3.2 Pollinator Community Composition Differs by Plant Flower Type

Flower morphology strongly filtered which pollinator groups visited each plant (Figure 2). Wild carrot attracted the most diverse community (bees = 63%, hoverflies = 16%, beetles = 11%, wasps = 10%). Comfrey and sainfoin were visited almost exclusively by bees (94% and 95% respectively). Within the bee community (Figure 2C), social bees — mainly bumblebees — dominated radish (73%) and comfrey (77%) visits, while solitary bees contributed more to carrot (59%). This sociality pattern directly motivates use of social bee abundance as the benchmark predictor for radish and comfrey in the JAE paper.

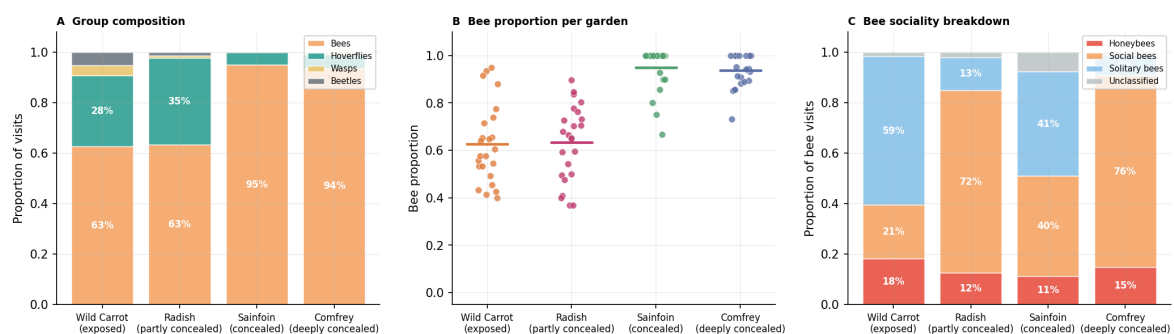


Figure 2. Pollinator community composition by plant species. (A) Mean proportion of visits by pollinator group. (B) Garden-level bee proportion (each point = one garden; line = mean). (C) Bee visit breakdown by sociality — honeybee, social (mainly *Bombus* spp.), solitary, unclassified.

3.3 Visitor Trait Profiles Already Differ by Plant Species

CWM bee trait profiles differed substantially by plant species (Figure 3). Bees visiting comfrey had the longest mean tongue lengths (mean = 8.66 mm vs. 2.23 mm for carrot), reflecting morphological filtering. Body size showed a similar gradient, with comfrey and sainfoin visited by larger bees (dominated by bumblebees). This pre-existing trait gradient makes the subsequent correlation results ecologically interpretable rather than spurious.

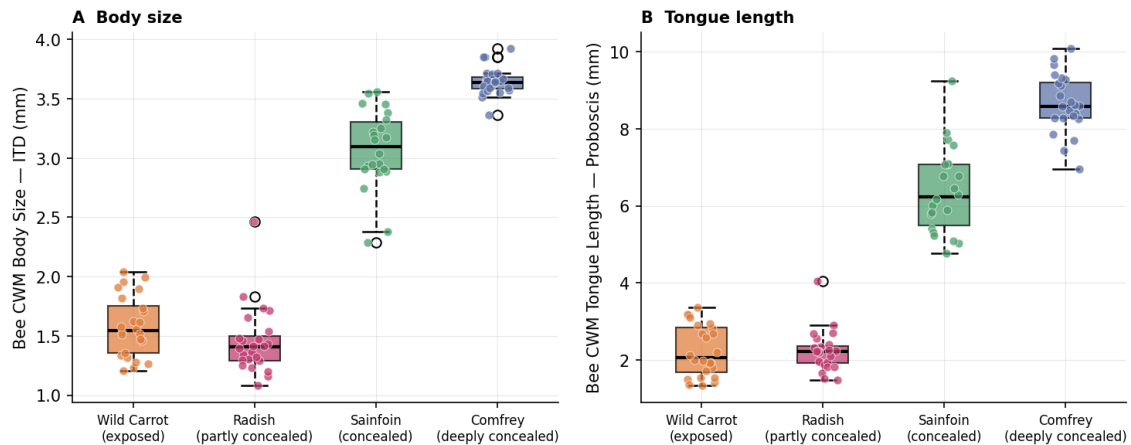


Figure 3. CWM bee trait profiles by plant species. (A) Mean bee body size (intertegular distance, mm). (B) Mean bee tongue length (proboscis length, mm). Each point = one garden. Horizontal line = mean.

3.4 Benchmark Replication: Focal Group Abundance

Testing the per-plant benchmark predictor from Reji Chacko et al. (2026) Figure S6, associations were in the expected positive direction for three of four plants (comfrey: $r = +0.12$; radish: $r = +0.15$; sainfoin: $r = +0.24$), though none reached significance at the garden level (Figure 4). This directional agreement validates my data aggregation pipeline. The null significance is consistent with lower statistical power compared to the original paper's Bayesian GLMMs, which controlled for random effects.

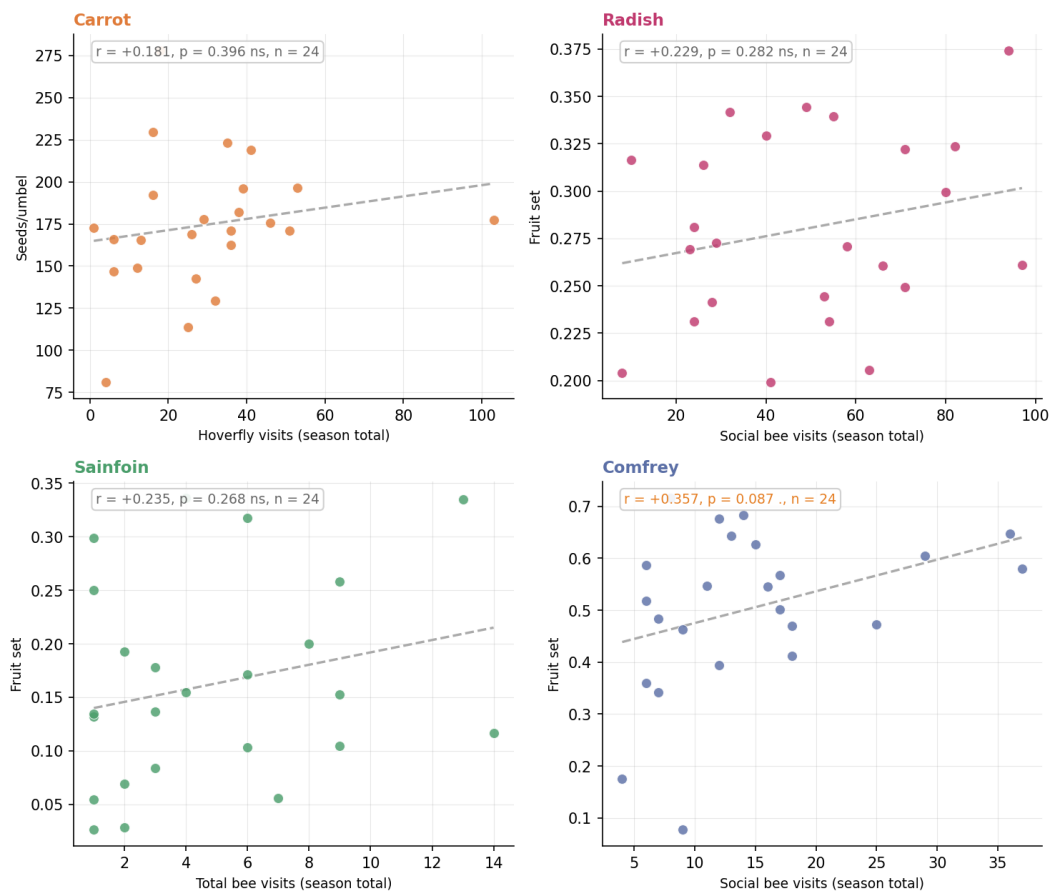


Figure 4. Benchmark replication (Reji Chacko et al., 2026, Figure S6). Focal group visits versus pollination success per plant. Each point = one garden. Solid line = significant ($p < 0.05$); dashed = not significant.

3.5 Bee Tongue Length Predicts Pollination Success in Specialist Flowers (RQ1)

CWM bee tongue length significantly predicted comfrey fruit set ($r = +0.45$, $p = 0.029$, $n = 24$; Figure 5D). Gardens visited by longer-tongued bee communities achieved higher comfrey fruit set — consistent with trait-matching theory: only long-tongued bees can reach the deeply concealed nectar and make legitimate contact with reproductive structures. Sainfoin showed a positive but non-significant trend ($r = +0.19$, $p = 0.37$). No significant association was found for wild carrot ($r = -0.26$, $p = 0.23$) — the open-nectar generalist where tongue length is ecologically irrelevant. The contrast between the comfrey result (significant) and the carrot result (null) is the core finding supporting flower-morphology-specific trait-matching.

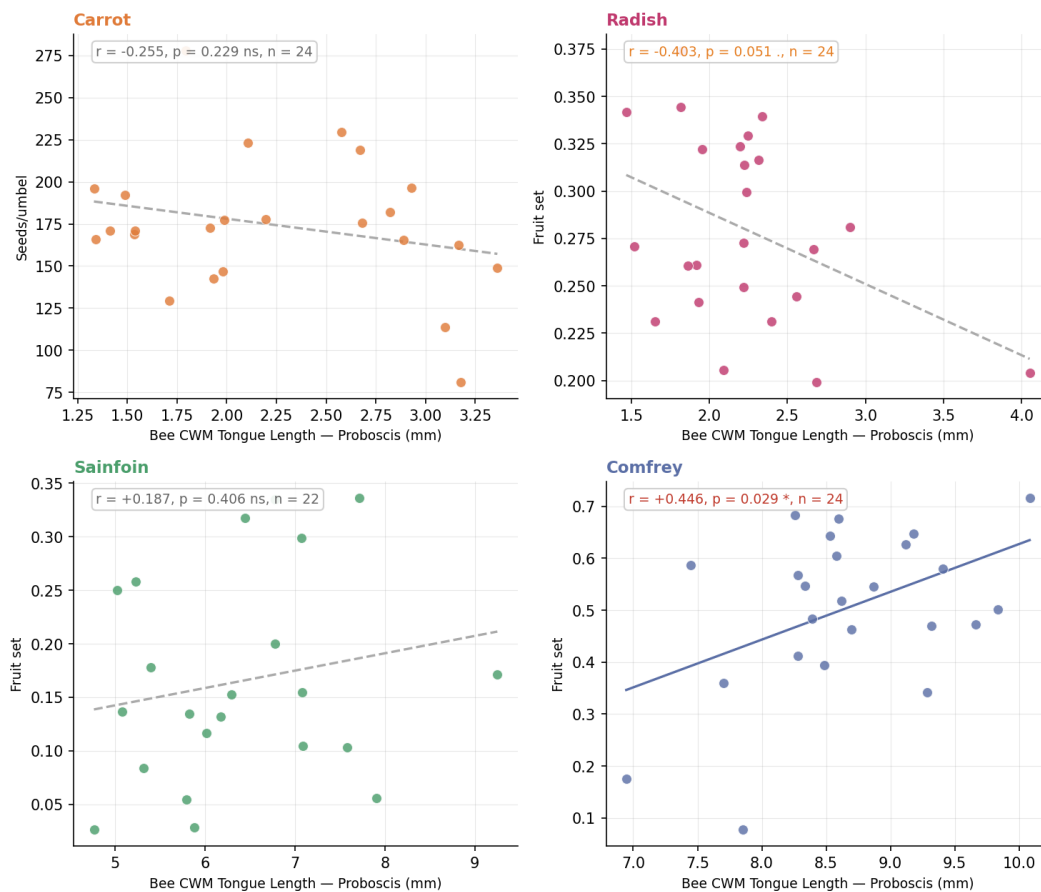


Figure 5. CWM bee tongue length (proboscis, mm) versus pollination success by plant species. Each point = one garden ($n = 24$). Solid line = $p < 0.05$; dashed = not significant. r and p values annotated in each panel.

3.6 Bee Body Size: An Exploratory Association with Radish Fruit Set (RQ2)

CWM bee body size showed an exploratory negative association with radish fruit set ($r = -0.47$, $p = 0.020$, $n = 24$; Figure 6B). No strong directional prediction was made for this association. One possible interpretation is that smaller bees are more effective at navigating radish's partly-concealed floral structure, though this requires experimental validation. Comfrey and sainfoin showed positive trends ($r = +0.30$ and $+0.21$ respectively), likely reflecting co-variation between body size and tongue length. All body size associations should be treated as exploratory.

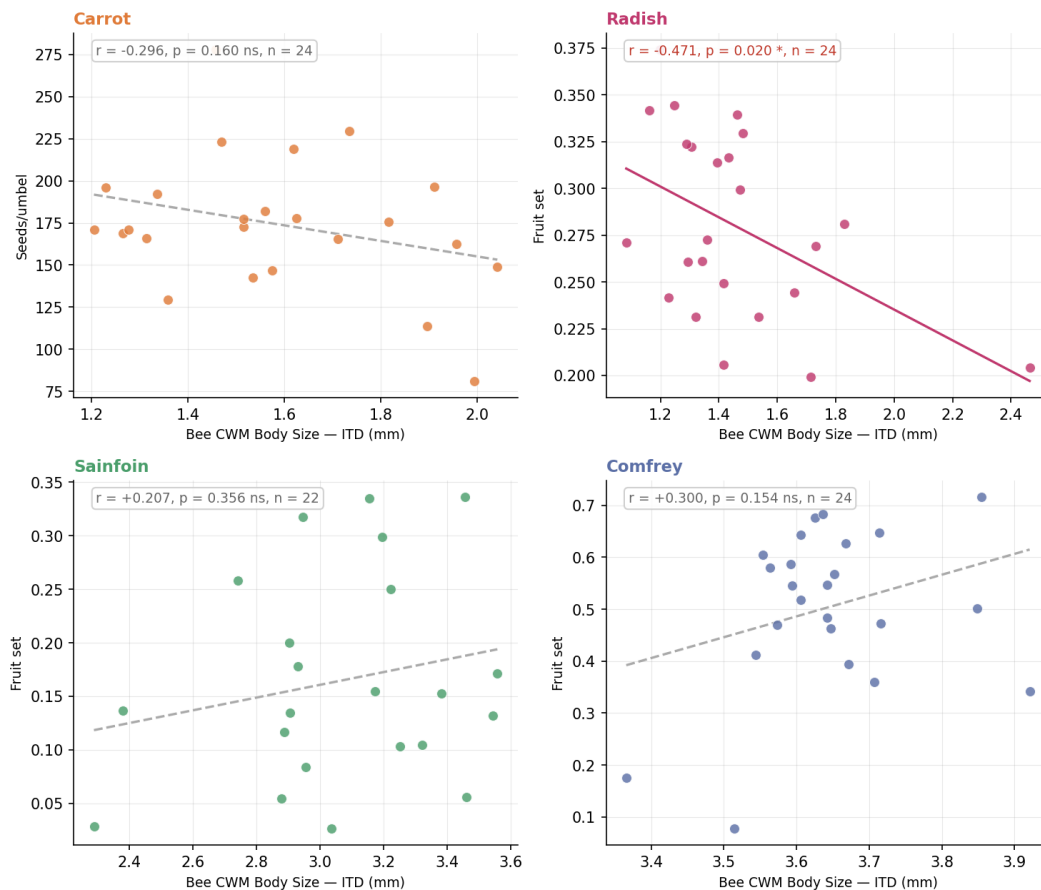


Figure 6. CWM bee body size (intertegular distance, mm) versus pollination success by plant species. Each point = one garden. Solid line = $p < 0.05$; dashed = not significant.

3.7 Social Bees Predict Comfrey Success Better Than Solitary Bees

For comfrey, social bee visits showed a stronger positive association with fruit set ($r = +0.37$, $p = 0.075$) than solitary bee visits ($r = +0.18$, $p = 0.39$; Figure 7, top row). This is consistent with bumblebees — the dominant social bee group — being large-bodied and long-tongued specialists capable of accessing comfrey's deeply concealed nectar. For radish, neither social bee visits ($r = +0.23$, $p = 0.28$) nor solitary bee visits ($r = -0.02$, $p = 0.92$) showed significant associations, suggesting total abundance of either bee type does not strongly drive radish success at the garden level.

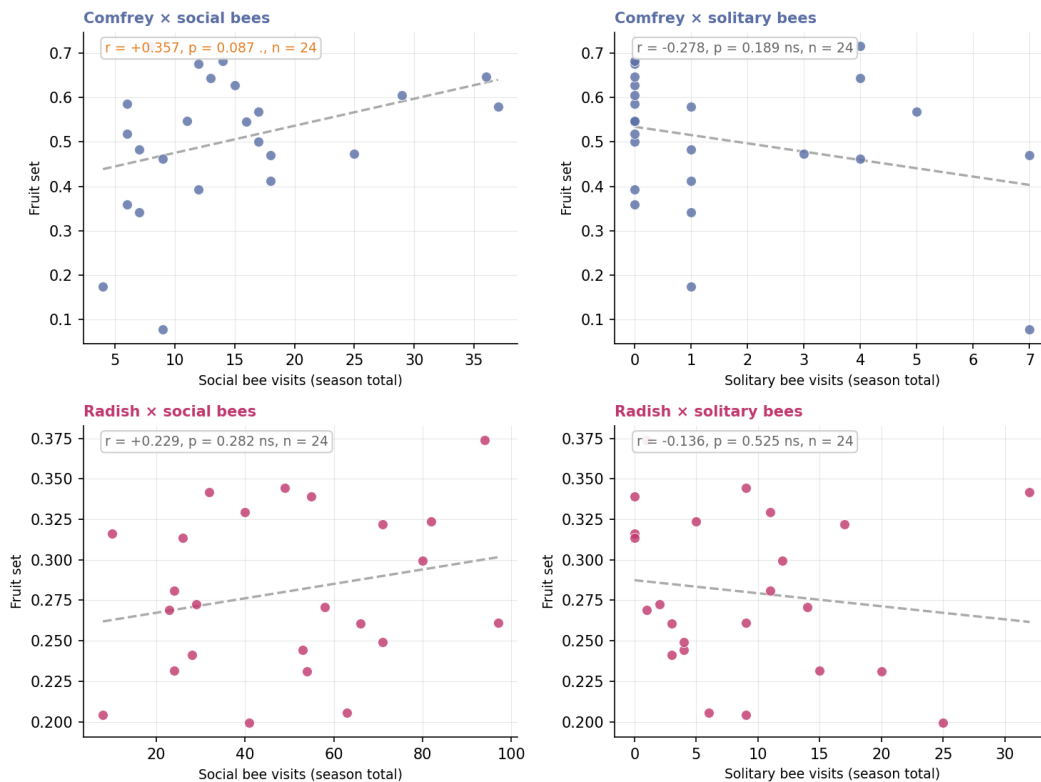


Figure 7. Social bee visits (left column) versus solitary bee visits (right column) as predictors of pollination success for comfrey (top row) and radish (bottom row). $n = 24$ gardens per panel.

3.8 Morning Visitation Predicts Radish Fruit Set (RQ3)

The proportion of daily visits occurring in the morning (09:00–12:00) was the strongest single predictor found in this study for radish fruit set ($r = +0.56$, $p = 0.004$, $n = 24$; Figure 8B), explaining 31% of variance. Gardens where a higher fraction of bee visits to radish occurred in the morning achieved significantly higher fruit set. A plausible mechanism is that radish stigma receptivity and pollen viability peak in morning hours, as observed in related Brassica species (Primack, 1985). No significant temporal association was found for carrot, sainfoin, or comfrey (all $p > 0.10$), making this result plant-specific rather than a general pattern.

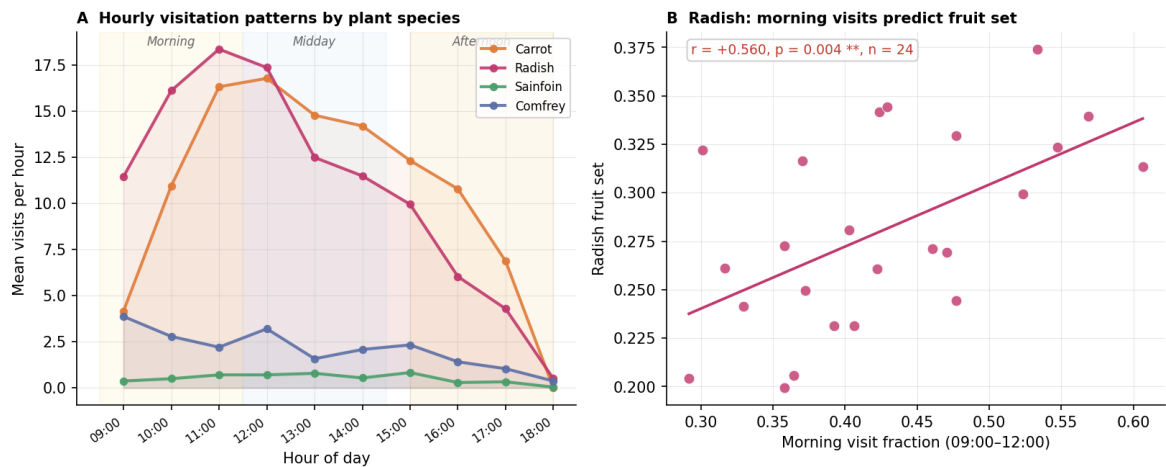


Figure 8. (A) Mean hourly flower visitation by plant species across the sampling day. Shaded bands = morning, midday, and afternoon time bins. (B) Morning visit fraction versus radish fruit set ($n = 24$ gardens). $r = +0.56$, $p = 0.004$.

3.9 Summary: Which Predictor Explains the Most Variance?

Table 1 presents R^2 values for all predictor types and all four plant species. For comfrey, CWM tongue length ($R^2 = 0.199^*$) exceeds both the JAE paper benchmark predictor ($R^2 = 0.014$) and total visit count ($R^2 = 0.014$). For radish, the morning ratio ($R^2 = 0.314^{**}$) substantially outperforms the benchmark ($R^2 = 0.023$) and bee visits ($R^2 = 0.000$). For carrot and sainfoin, no predictor explains more than 16% of variance. Trait composition and temporal patterns contain predictive information beyond group abundance, but only for plants where flower morphology creates specificity in visitor requirements.

	Wild Carrot (exposed)	Radish (partly concealed)	Sainfoin (concealed)	Comfrey (deeply concealed)
Benchmark (JAE Fig S6)	0.033	0.052	0.055	0.128.
Abundance — total visits	0.006	0.029	0.057	0.014
Abundance — bee visits	0.086	0.000	0.055	0.011
Sociality — social bee visits	0.057	0.052	0.016	0.128.
Sociality — solitary bee visits	0.019	0.019	0.089	0.077
Trait — bee tongue length (CWM)	0.065	0.162.	0.035	0.199*
Trait — bee body size (CWM)	0.088	0.222*	0.043	0.090
Temporal — morning ratio	0.043	0.314**	0.155.	0.025
Temporal — evenness	0.000	0.052	0.051	0.000

R^2 values shown. Significance: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$. $p < 0.10$. Dark red/orange = significant.

Table 1. Proportion of variance explained (R^2) by predictor category and plant species. Darker cells = higher significance (orange $p < 0.05$; red $p < 0.01$; dark red $p < 0.001$; yellow $p < 0.10$).

4. Discussion

4.1 Benchmark Associations Replicate the JAE Paper Direction

The focal group benchmark associations in Figure 4 match the positive directions reported by Reji Chacko et al. (2026) in Figure S6 for three of four plant species. The null significance at the garden level is expected: the original paper's Bayesian GLMMs controlled for sampling effort, garden, sampling date, and time window as crossed random effects — none of which my analysis replicates. The directional agreement nonetheless validates my data aggregation pipeline and confirms that the key abundance signal is recoverable at the garden level with simpler methods.

4.2 Trait-Matching: Tongue Length Predicts Success on Specialist Flowers

The significant positive association between CWM bee tongue length and comfrey fruit set ($r = +0.45$, $p = 0.029$) is the primary novel finding of this study. The JAE paper demonstrated that traits respond to environmental gradients but did not test whether trait composition predicts pollination outcomes. My analysis shows it does, at least for the most specialised flower in the experiment. This is consistent with Armbruster's (2017) specialisation continuum: co-specialisation between long-tongued bees and euphilous flowers creates a more reliable pollination mutualism, and communities with higher mean tongue length realise this mutualism more fully. The null result for wild carrot — the open generalist flower — further supports this interpretation: where any visitor can access the reward, trait composition is irrelevant to reproductive success.

The exploratory negative association between CWM body size and radish fruit set ($r = -0.47$, $p = 0.020$) is intriguing but should be treated with caution. Gardens dominated by smaller bee visitors may achieve more efficient pollination of radish's partly-concealed flowers, possibly because smaller bees make more contact with reproductive structures. However, this interpretation lacks direct mechanistic support, and the correlation may reflect confounding variables such as community composition. Future work with pollen deposition measurements would be needed to test this mechanism directly.

4.3 Temporal Patterns: A Novel Predictive Dimension for Radish

The finding that morning visit fraction predicts radish fruit set ($r = +0.56$, $p = 0.004$) — explaining 31% of variance and outperforming all abundance and trait predictors for that plant — is the most unexpected result of this study. This fills a gap explicitly noted by the data publication, which identified the hourly temporal resolution as an opportunity for future analysis. A plausible biological mechanism involves diurnal variation in flower receptivity: many Brassicaceae relatives of radish show peak stigma receptivity and pollen viability in the morning hours (Primack, 1985), so morning-concentrated visitation may be more effective per contact. This mechanism cannot be confirmed from this dataset, which lacks flower receptivity measurements. Importantly, the temporal result is plant-specific: no temporal predictor was significant for carrot, sainfoin, or comfrey, reducing the risk that it is a statistical artefact.

4.4 Limitations and Future Directions

Several important limitations constrain interpretation. First, $n = 24$ gardens provides limited statistical power, and all correlations should be treated as hypothesis-generating. The JAE paper's Bayesian GLMMs with crossed random effects provide a more rigorous analytical framework. Second, I do not control for landscape-scale urban densification or local floral richness — two covariates that significantly drive pollinator abundance in the original analysis. Some correlations I report may partially reflect these unmeasured gradients. Third, the sainfoin caution flag is a proxy based on flower count and does not reproduce the exact three-garden exclusion in the JAE paper.

Future analyses could: (1) apply linear mixed-effects models with densification and floral richness as covariates; (2) test functional diversity metrics (FDis, Rao Q) alongside CWM values; (3) replicate the temporal analysis across the BetterGardens network's other Swiss cities (Lausanne, Bern); and (4) couple trait data with single-visit pollen deposition measurements to directly test pollinator effectiveness hypotheses.

5. Conclusion

This study demonstrates that pollinator functional trait composition and temporal visit distribution predict garden-level pollination success in plant-species-specific ways that abundance measures alone do not capture. CWM bee tongue length significantly predicted comfrey fruit set ($r = +0.45$, $p = 0.029$), consistent with trait-matching theory for a deeply euphilous flower. Morning visit fraction was the strongest predictor found for radish fruit set ($r = +0.56$, $p = 0.004$) — a temporal dimension never previously tested on this dataset. Neither trait nor temporal predictor was significant for wild carrot, consistent with its open generalist flower morphology.

These findings address three gaps relative to Reji Chacko et al. (2026): (1) I test pollinator traits as predictors of success rather than responses to environment; (2) I introduce temporal visitation distribution as a novel predictor dimension; (3) I disaggregate bee visits by sociality to enable direct comparison with the JAE paper benchmark. The trait and temporal predictors outperform the abundance-based benchmarks in R^2 terms for both comfrey and radish, suggesting that the identity of pollinators — who visits and when — matters alongside quantity.

The results are exploratory, constrained by $n = 24$ gardens and the absence of landscape-level covariates. They nonetheless provide testable hypotheses for future experimental work on trait-mediated pollination services in urban environments. The core message complements the JAE paper's broader conclusion — that no single metric fully characterises pollination service — by adding that functional trait composition and daily timing of visits each contribute predictive information beyond pollinator abundance alone.

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- Dataset:** Reji Chacko, M., Moretti, M., & Frey, D. J. (2025). Comprehensive dataset of pollinator diversity and visitation rates [Dataset]. EnviDat. <https://doi.org/10.16904/envidat.676>
- Analysis tools:** Python 3.10; pandas; NumPy; SciPy; Matplotlib. All code available in accompanying Jupyter notebooks. AI tools were used to assist with literature review, data analysis, and narrowing research questions.